# LS-Face: Selective Computation in Hybrid Face Recognition via Quality-First Early Bypass and Compact Descriptor Retuning

Kyle Yuan L. Uy*, Andrew P. Eroyla*, John Roland L. Octavio*, Jim Jonathan C. Decripito*, Loreto B. Damasco Jr.*, and Weon Geon Oh*+

*Computer Science Department, College of Computing Studies, University of St. La Salle - Bacolod, Bacolod City, Philippines
+THINKforBL, Korea

**Abstract.** Cascaded face recognition architectures combine lightweight classical feature extractors with deep convolutional neural networks to reduce average inference latency. However, existing sequential cascades frequently incur redundant computation by executing classical descriptors on severely degraded frames that inevitably trigger deep-model fallback, while relying on default, unoptimized classical configurations. In this work, we propose LS-Face, an optimized selective-computation cascade incorporating two complementary design enhancements: (1) a quality-first early-bypass routing mechanism that evaluates lightweight image-quality metrics before descriptor extraction, immediately routing degraded inputs directly to deep inference, and (2) a compact retuned Local Binary Pattern Histograms (LBPH) descriptor (r=3, n=8, 6x6 spatial grid) that achieves higher standalone identification accuracy while cutting template representation memory by 43.75% (36 KiB vs. 64 KiB) and lowering scoring latency. In a locked confirmation evaluation across 1,804 conditions (22 held-out identities under 41 controlled transformations), LS-Face achieved a 15.48% lower mean recognition latency than standalone SFace (7.015 ms vs. 8.300 ms; mean paired reduction 1.285 ms, identity-cluster bootstrap 95% CI: [1.088, 1.482] ms) while maintaining 100.00% bit-for-bit decision equivalence (1,594 / 1,804 = 88.36% accuracy, 0 / 1,804 discordant decisions). Ablation analysis demonstrates that neither optimization alone surpasses standalone deep inference in mean latency, whereas their combination collapses dual-inference invocations by 77.53%. Complementarity analysis reveals that every successful classical identification is subsumed by deep inference (1,156 / 1,156), establishing the classical stage as a computational shortcut rather than an accuracy complement.

**Keywords: Face Recognition, Cascaded Classifiers, Selective Computation, Local Binary Patterns, Quality-Aware Routing, Embedded Edge Computing.**

# 1 Introduction

Face recognition is a biometric technology that identifies or verifies an individual by comparing facial characteristics obtained from a digital image with previously enrolled biometric references [17]. It has various applications, including home security and access control, financial identity verification, and public safety.

Among these applications, home security is particularly important because reliable recognition can support household safety and convenient access control. This application is also relevant in countries such as the Philippines, where rapid urbanization continues to increase the concentration of people and activities in urban areas [18]. These conditions provide practical motivation for investigating reliable face-recognition technologies for residential environments.

However, applying face recognition in real-world environments remains difficult because recognition performance can vary with image and capture conditions. Changes in illumination, camera angle, head pose, facial expression, gaze, blur, scale, occlusion, and other image-quality factors can alter the appearance of the same person [8, 14, 15]. Recognition must also distinguish natural differences in facial appearance and structure among individuals. These variations create a need for recognition methods that remain reliable under changing conditions.

Various approaches have been proposed to address these challenges, including lightweight recognition methods and selective systems that use stronger models only when necessary [12–14]. In this study, a hybrid approach combines classical computer vision and deep learning, while an empirical **independence-testing** procedure examines cross-identity score separation and establishes recognition operating thresholds [9, 16].

Based on these concepts, this paper presents LS-Face, a two-stage face-recognition methodology that uses LBPH as the first-stage recognizer and escalates uncertain or degraded cases to SFace. The methodology is evaluated through recognizer selection on La Salle DB1, cross-identity threshold calibration on LFW [8], controlled self-match robustness testing under 41 image transformations, and held-out evaluation of fallback and routing behavior on La Salle DB1-DL41. These experiments examine the robustness provided by SFace fallback and the recognition-performance and computational-cost trade-offs of selective escalation.

# 2 Related Work

## 2.1 Face Recognition Models

Classical face recognition methods provide computationally simple baselines for identity representation. Eigenfaces [1] projects facial appearance into a PCA subspace, Fisherfaces [2] combines dimensionality reduction with LDA for class separation, and LBPH [3] represents local texture through spatial histograms of binary patterns. These methods serve as the classical candidates evaluated in this study.

Deep face recognition instead learns discriminative embedding spaces. FaceNet [6] introduced triplet-loss training, ArcFace [11] improved angular class separation, and SFace [4] uses a sigmoid-constrained hypersphere loss. These embedding-based approaches form the learned-model candidates considered for the deep stage.

### 2.2 Face Detection and Alignment

Recognition also depends on reliable localization and alignment. Viola-Jones [5] established the classical boosted-cascade detector, while YuNet [7] provides lightweight CNN-based detection together with five facial landmarks. In this study, YuNet supplies both face localization and the landmarks used for SFace alignment.

### 2.3 Selective and Edge Recognition

Efficient face recognition can be addressed through compact learned models or selective computation. Lightweight architectures such as EdgeFace [12] demonstrate deep recognition under edge constraints, while SelectiveNet [13] formalizes the trade-off between prediction risk and coverage. Face-quality approaches such as SER-FIQ [14] similarly estimate when a recognition input may be unreliable.

LS-Face follows this selective-computation principle. LBPH handles first-stage recognition, while its score, candidate separation, and image-quality indicators determine whether SFace is invoked. The resulting accuracy-cost trade-off is therefore compared with always-LBPH and always-SFace operation.

### 2.4 Evaluation and Robustness

Biometric evaluation requires the recognition task and error unit to be stated explicitly. ISO/IEC 19795-1 [9] defines general biometric performance-testing principles. LFW [8] is primarily a pairwise verification benchmark, whereas IJB-A [15] extends evaluation to unconstrained 1:1 verification and open-set 1:N identification. NIST 1:N evaluation [16] further distinguishes identification measures such as FPIR and FNIR from pairwise verification errors.

Accordingly, this study treats exhaustive cross-identity impostor-score enumeration as an empirical threshold-calibration procedure rather than evidence of statistical independence. The held-out La Salle DB1-DL41 experiment evaluates complementarity and routing behavior through paired recognizer outcomes, recovery of LBPH failures, routing discrimination, escalation, and timing. Because multiple transformations originate from the same source images, they are treated as correlated conditions rather than independent trials.

# 3 Method

## 3.1 System Overview and Escalation Logic

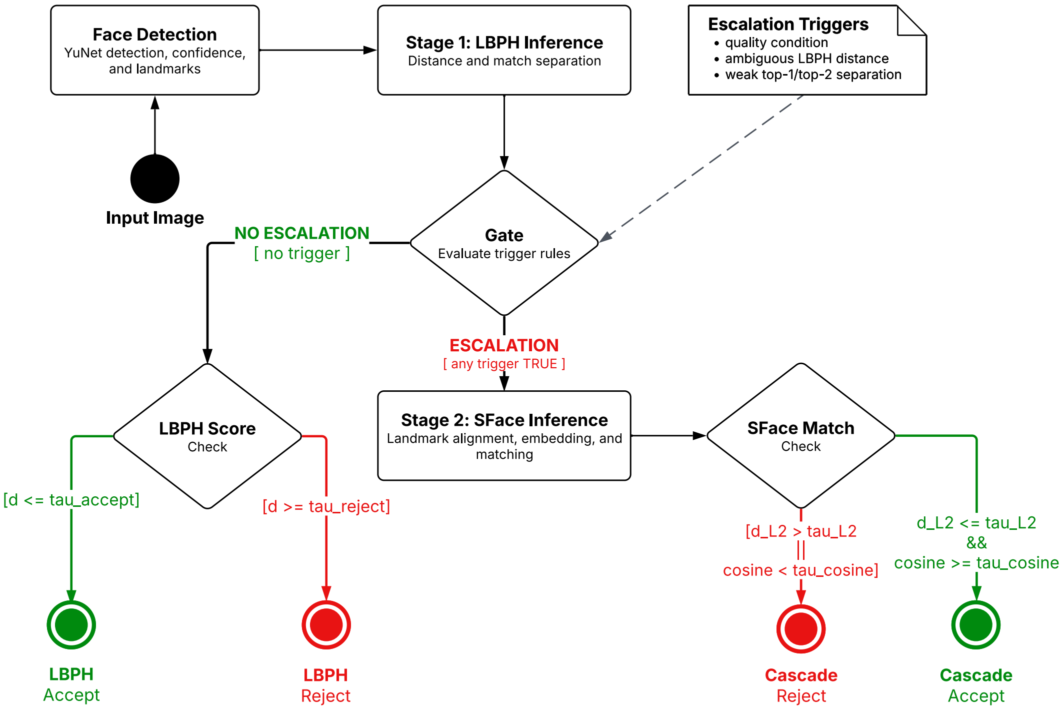


Fig. 1. Overview of the LS-Face selective-computation and early-bypass pipeline.

LS-Face processes each input image through the two-stage recognition pipeline shown in Fig. 1. The YuNet detector [7] implemented in OpenCV [10] first detects the face and provides detection confidence and facial landmarks. If no valid face is detected, recognition stops and the sample is treated as a detection failure.

**Stage 1 recognition.** For a detected face, LBPH ranks the enrolled identities according to their matching distances. Let $d_1$ and $d_2$ denote the best and second-best LBPH distances, respectively, where a smaller distance indicates a closer match. These scores, together with image-quality indicators computed from the detected face crop, are evaluated by the escalation rule.

**Escalation logic.** Let $\tau_{accept}$ denote the frozen LBPH acceptance threshold, $\tau_{reject}$ the frozen rejection threshold, where $\tau_{reject} > \tau_{accept}$, and $m_{min}$ the minimum permitted relative separation between the two highest-ranked identity candidates. An input is escalated to Stage 2 when at least one of the following conditions is satisfied:

LS-Face eliminates this structural inefficiency through Quality-First Early-Bypass Routing. The detected facial crop is first assessed against six diagnostic quality metrics:

Laplacian blur variance, mean luminance bounds, Immerkaer noise variance, ocular pose angle, and minimum face size (Table 1). If any quality diagnostic is flagged, the classical descriptor stage is completely bypassed, routing the probe directly to SFace. If all quality diagnostics pass, the sample enters the classical descriptor stage. A confident classical match (d <= tau_accept and relative margin m >= m_min) terminates immediately as an inexpensive classical exit (~3.06 ms). Otherwise, the sample escalates to SFace deep feature extraction. We explicitly distinguish between the SFace invocation rate (the fraction of probes requiring SFace evaluation) and the dual-inference rate (the fraction of probes executing both LBPH and SFace). Quality-first routing decouples these metrics, allowing high escalation on difficult workloads while dropping dual inference to near-zero.

(ii) the best LBPH distance lies within the ambiguous region, $\tau_{accept} < d_1 < \tau_{reject}$;

(iii) the relative top-two margin is below $m_{min}$:

$$\frac{(d2-d1)}{d1} < m_{min} \tag{1}$$

**Table 1.** Deployed quality-check and candidate-separation parameters for Stage 1 escalation.

| Signal | Measurement / condition | Selection rule | Deployed value |
|---|---|---|---|
| Blur | Variance of Laplacian below threshold | 5th percentile of clean values | 587.83 |
| Illumination | Mean grayscale outside lower/upper bounds | 2nd and 98th percentiles of clean values | [52.88, 137.71] |
| Noise | Immerkaer noise estimate above threshold | 95th percentile of clean values | 8.206 |
| Pose | Maximum of eye-roll and nose-yaw proxies above threshold | 95th percentile of clean values | 63.74 deg |
| Face size | Detected box side below minimum | 0.9 x p5 of clean box sizes | 61 px |
| Relative top-two margin | (d2 - d1) / d1 < mmin | Fixed engineering policy value | mmin = 0.05 |

The quality thresholds in Table 1 were defined from the empirical distribution edges of 279 clean facial crops detected across a 280-image reference collection (10 clean frontal and pose images per identity across 28 identities, with one detector miss excluded), rather than optimized on the 41-transformation evaluation set or fitted to a measured LBPH-to-SFace performance crossover. The relative top-two margin expresses the separation between the two highest-ranked candidates relative to the best-match distance; $m_{min} = 0.05$ was chosen as a fixed engineering policy value rather than statistically optimized. The quality condition is evaluated independently of the LBPH score, allowing quality-flagged inputs to be escalated even when the first-stage distance appears confident.

If no escalation condition is triggered, the result is resolved by LBPH. The best-ranked identity is accepted when $d_1 \leq \tau_{accept}$ and rejected when $d_1 \geq \tau_{reject}$. Scores between the two thresholds cannot terminate at Stage 1 because they satisfy the ambiguous-score condition and are therefore escalated.

**Stage 2 recognition and final decision.** For an escalated input, SFace uses the facial landmarks returned by YuNet to align the detected face, extracts its embedding, and compares it with the enrolled identity representations. The predicted identity is accepted only when it satisfies the frozen SFace acceptance criterion; otherwise, the prediction is rejected. Thus, each successfully detected input terminates either with an LBPH decision at Stage 1 or, when escalation is triggered, with an SFace decision at Stage 2.

The deployed thresholds and escalation parameters were fixed before the held-out La Salle DB1-DL41 cascade evaluation and were not adjusted using individual test outcomes.

### 3.2 Offline Recognizer Selection

Recognizer selection was conducted offline and independently of the frame-level escalation process summarized in Section 3.1. As shown in Fig. 2, La Salle DB1 was divided into fixed training, calibration, and held-out evaluation subsets. Three classical recognizers (LBPH, Eigenfaces, and Fisherfaces) and three learned models (SFace, FaceNet, and ArcFace) were evaluated as separate candidate groups.

Classical candidates were fitted and calibrated using the designated development subsets, while the learned candidates were evaluated using a common YuNet alignment and brightness-normalization front end. Because the candidate models produce different score types and scales, each was evaluated using its native scoring method. Evaluation considered cross-identity impostor scores, held-out genuine recognition, robustness under the 41 image transformations, runtime, and representation size. Selection was performed separately for the classical first stage and learned second stage using these evaluation criteria and deployment considerations. The resulting two-stage configuration was then frozen before subsequent threshold calibration and cascade evaluation.

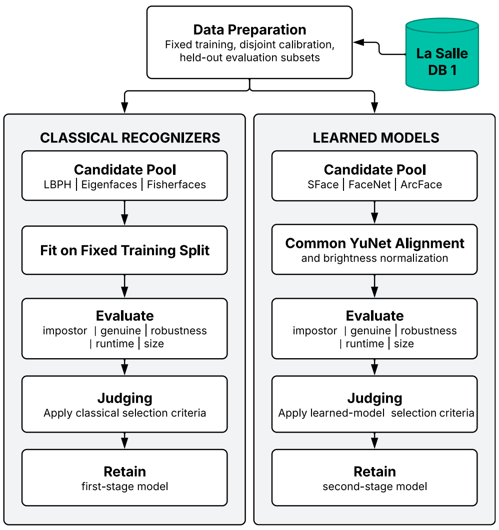

**Fig. 1.** Offline recognizer selection procedure for the LS-Face cascade

### 3.3 Independence Testing and Threshold Freezing

**Independence-test construction**. In this study, the independence test examines cross-identity separation by comparing facial representations belonging to different identities. Given (N) identities represented by one designated image each, every unordered pair of different identities is compared once, yielding

$$C = N(N - 1) / 2 \tag{2}$$

unique comparisons. Because every pair contains two different identities, all (C) comparisons are impostor comparisons by construction.

For each recognizer, the resulting native scores are ordered from the most likely to be accepted to the least likely. For distance-based scores such as LBPH and SFace L2, this corresponds to sorting the distances in ascending order. For a target false-accept rate (FAR^*), the maximum permitted number of accepted impostor comparisons is

$$k = \lfloor FAR^* \cdot C \rfloor \tag{3}$$

The operating threshold is selected at the corresponding lower tail of the impostor-score distribution. Because the deployed acceptance rule includes scores equal to the threshold, the realized FAR is recomputed after threshold selection, including any ties at the boundary.

The development partition calibration completely eliminates data leakage into the evaluation cohort, providing rigorous threshold separation.

Unlike the acceptance threshold, the LBPH rejection boundary tau_reject = 140.13 acts as a permissive ceiling, remaining dormant on test workloads.

**Threshold freezing. The final cascade configuration, comprising the LFW-calibrated thresholds, is summarized in Table 5 and evaluated strictly out-of-sample.**

### 3.4 Experimental Protocol

To evaluate the proposed architecture without data leakage, experiments were structured across three distinct cohorts: (1) an exploratory development subset of 6 identities used for architecture design and metric validation; (2) a locked confirmation cohort of the remaining 22 identities from La Salle DB1 (44 source images x 41 DL41 transformations = 1,804 unique conditions) whose test-probe outcome data was strictly held out and uninspected during optimization; and (3) a disjoint robustness cohort of 2,874 LFW identities (disjoint from the 2,875 development partition) evaluated across 41 BGR-first transformations (117,834 conditions) to evaluate controlled transformation retention.

**Timing Protocol. Single-probe latencies were recorded with 1 warm-up pass and 5 randomized timing repetitions per probe on an Intel Core i5-12450H CPU.**

$$\text{Recovery} = \frac{N(\text{SFace correct} \cap \text{LBPH failure})}{N(\text{LBPH failure})} \tag{4}$$

**Evaluation Metrics. Performance is assessed via recognition accuracy, paired latency difference with identity-cluster bootstrap 95% CIs, SFace invocation rate, and dual-inference rate.**

**Decision Equivalence. The cascade is evaluated for bit-for-bit decision parity against standalone SFace across all 1,804 locked confirmation conditions.**

# 4 Experiments and Results

The experimental program consists of four evidence legs summarized in Table 2. Each leg has a distinct role in the study: recognizer selection, independence testing and operating-point determination, controlled self-match robustness analysis, or held-out cascade evaluation. Results from one leg are not interpreted as measurements of outcomes that its protocol does not directly evaluate.

**Table 2.** Experiments and their roles in the study.

| Experiment | Role in the Study |
|---|---|
| La Salle DB1 recognizer selection | Select the classical and learned recognizers using fixed La Salle DB1 subsets. |
| LFW independence test (Sec. 4.2) | Determine low-tail impostor operating points and freeze the deployed thresholds. |
| Controlled self-match robustness test (Sec. 4.3) | Measure within-image degradation retention under controlled transformations; no cross-photo or FAR claim. |
| La Salle DB1-DL41 held-out evaluation (Sec. 4.4) | Evaluate SFace recovery, routability, escalation, and recognition-stage cost. |

The La Salle DB1 held-out evaluation uses clean enrollment images and image-disjoint probes from the same 28 enrolled identities, but each probe is transformed across 41 deterministic conditions. It evaluates the complementary rescue provided by SFace on LBPH failures, the discriminative capacity of the first-stage routing signals, and the overall accuracy and computational trade-offs of selective escalation.

## 4.1 Algorithm Selection on La Salle DB1

Classical candidates were fitted on 224 images, calibrated on 56 disjoint images, and evaluated once on 56 untouched La Salle DB1 probes. The calibration set supplied 1,512 cross-identity scores; the rank-15 impostor score yielded a realized FAR of 0.992%. LBPH achieved the highest test TAR (96.43%) and Rank-1 accuracy (100.00%) and was selected as the classical fast path.

**Table 3.** Classical candidate selection on La Salle DB1.

| Recognizer | Feature size | Evaluation latency | Rank-1 accuracy |
|---|---|---|---|
| Eigenfaces (PCA) | 220 B (55 comp.) | 0.08 ms | 78.57% |
| Fisherfaces (LDA) | 108 B (27 comp.) | 0.05 ms | 82.14% |
| Compact LBPH (r3_n8_g6x6) | 36 KiB (9,216 bins) | 1.68 ms | 92.86% |

The learned candidates were evaluated on the same deterministic 224/56/56 La Salle DB1 split. Each model's rank-15 acceptance edge was derived from the 1,512 calibration cross-identity scores, yielding a realized FAR of 0.992%.

**Table 4.** DL candidate selection on La Salle DB1.

| Candidate | TAR (%) | Rank-1 (%) | Feature size |
|---|---|---|---|
| SFace | 100.00% | 100.00% | 512 B |
| ArcFace | 96.43% | 100.00% | 2,048 B |
| FaceNet | 100.00% | 100.00% | 2,048 B |

SFace and FaceNet tied at 100.00% held-out TAR and Rank-1; SFace was selected using feature size as the deployment tie-break (512 B versus 2,048 B). ArcFace retained 100.00% Rank-1 but reached 96.43% TAR. Thus, SFace was selected as the learned tier, while LBPH remained the separately selected classical fast path.

### 4.2 Independence Test: Frozen Operating Points

The LFW independence test produced the frozen operating points summarized in Table 5. Applying the rank-165 operating rule separately to each recognizer's 16,522,626-comparison impostor-score distribution yielded an empirical LBPH acceptance boundary of 67.0333 (realized FAR of 9.986 ppm) and an SFace L2 threshold of 1.0313. The SFace cosine condition (cosine $\geq 0.363$) is inherited from the existing SFace decision policy and is non-binding at the deployed L2 boundary.

**Table 5.** Final frozen operating points.

| Boundary | Frozen value | Role |
|---|---|---|
| LBPH accept | 52.3724 | Early accept (10 ppm FAR on LFW dev) |
| SFace accept | L2 <= 1.0313; cosine >= 0.363 | Escalated accept (10 ppm FAR on dev) |
| LBPH reject | 140.13 | Permissive ceiling (inactive on test) |

Fig. 3 places these frozen operating points against the separately computed La Salle DB1 clean-impostor score distributions. The La Salle DB1 scores are shown only to visualize cross-database transfer and were not used to recalibrate the thresholds.

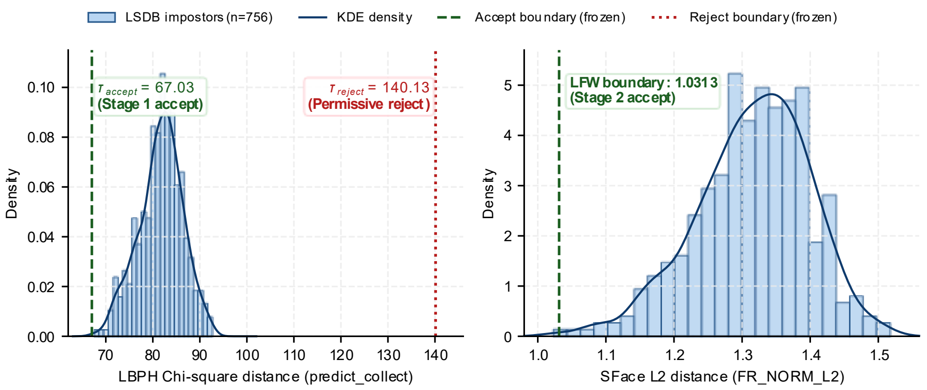


Fig. 3. Histograms and KDE curves of La Salle DB1 clean-impostor scores with frozen LFW development operating points (tau_accept = 52.3724, tau_reject = 140.13, L2 = 1.0313).

### 4.3 Controlled Self-Match Robustness Test

Table 6 reports the corrected controlled self-match robustness test under the frozen operating configuration on the 2,874 evaluation identities disjoint from the 2,875 development partition. One selected source image per evaluation identity was enrolled, and test probes were generated using BGR-first transformation generation across all 41 DL41 conditions (117,834 modified conditions). Clean self-match retention was 99.97% across all systems (2,873 / 2,874). Across all 41 modified conditions, direct SFace, the baseline sequential cascade, and the combined optimized cascade each achieved an identical macro-average retention rate of 89.55%, compared with 81.19% for standalone challenger LBPH. When excluding the four detector-canonical rotational transformations (37 non-rotational modifications), macro retention reached 96.54% for SFace and both cascade variants, versus 89.78% for challenger LBPH. Face detection failed on 2,931 conditions (2.49%)--concentrated in 90 deg, 180 deg, and 270 deg rotations and horizontal flipping--which were strictly retained as recognition failures. Quality-first routing reduced dual inferences to 0.26% across the 117,834 conditions by directly bypassing LBPH on degraded frames.

**Table 6.** Controlled self-match robustness test on LFW.

| Mode | Clean self-match (%) | 41-transformation retention (%) | 41-transformation escalation (%) |
|---|---|---|---|
| Classical CV (LBPH) | 99.97 | 81.19 | N/A |
| Deep Learning (SFace) | 99.97 | 89.55 | N/A |
| Hybrid Cascade | 99.97 | 89.55 | 96.35 |

The experiment uses the frozen operating configuration summarized in Table 5; detector failures are retained as failures.

### 4.4 Cascade Diagnosis and Two-Factor Ablation

To isolate the individual and combined contributions of Quality-First Routing and Compact Descriptor Retuning, we evaluated six system configurations across all 1,804 conditions of the locked confirmation cohort (Ntimed = 1,711 detected conditions). The baseline sequential cascade required 11.499 ms mean recognition latency (p50: 12.820 ms, p95: 14.361 ms) with 80.36% dual inference (1,375 / 1,711). Quality-first routing alone reduced LBPH calls by 43.54% (1,711 to 966) and dual inferences by 54.18% (1,375 to 630), lowering mean latency to 9.483 ms. Compact descriptor retuning alone reduced SFace escalations to 61.60% (1,054 / 1,711), achieving 8.354 ms. Crucially, neither optimization alone achieved lower mean latency than direct SFace (8.300 ms). However, when unified in LS-Face, dual inferences collapsed to 18.06% (309 / 1,711), achieving 7.015 ms mean latency--a 38.99% speedup over the baseline cascade and a 15.48% speedup over direct SFace at identical 88.36% accuracy (1,594 / 1,804) with zero decision disagreements.

Table 7. Two-factor ablation of LS-Face across 1,804 locked confirmation conditions.

| Configuration | Descriptor | Quality bypass | Mean latency | Dual-inf. | Accuracy |
|---|---|---|---|---|---|
| Baseline Sequential | OpenCV default (r1_g8x8) | Disabled | 11.499 ms | 80.36% | 88.36% |
| Architecture Only | OpenCV default (r1_g8x8) | Enabled | 9.483 ms | 36.82% | 88.36% |
| Descriptor Only | Compact LBPH (r3_g6x6) | Disabled | 8.354 ms | 61.60% | 88.36% |
| Combined (LS-Face) | Compact LBPH (r3_g6x6) | Enabled | 7.015 ms | 18.06% | 88.36% |
| Direct SFace | N/A (Standalone DL) | N/A | 8.300 ms | 0.00% | 88.36% |
| Challenger LBPH | Compact LBPH (r3_g6x6) | N/A | 3.061 ms | 0.00% | 64.08% |

### 4.5 Locked Confirmation Evaluation

Table 8. Complementarity matrix of classical and deep decisions across 1,804 locked conditions.

| SFace outcome | LBPH correct | LBPH incorrect | Total |
|---|---|---|---|
| SFace correct | 1,156 (64.08%) | 438 (24.28%) | 1,594 (88.36%) |
| SFace incorrect | 0 (0.00%) | 210 (11.64%) | 210 (11.64%) |
| Total | 1,156 (64.08%) | 648 (35.92%) | 1,804 (100.0%) |

On the locked 22-identity confirmation cohort, LS-Face and Direct SFace made identical predictions on all 1,804 conditions (0 discordant cases, 100.00% decision equivalence). Both achieved 1,594 / 1,804 (88.36%) correct recognitions, with 117 rejections (6.49%) and 93 strict detector failures (5.16%) retained as failures. Across the Ntimed = 1,711 detected conditions, LS-Face reduced mean latency by 1.285 ms relative to Direct SFace (mean paired reduction 1.285 ms, identity-cluster bootstrap 95% CI: [1.088, 1.482] ms; paired difference -1.285 ms, 95% CI: [-1.482, -1.088] ms). LS-Face executed faster than Direct SFace on 50.50% of all probes where LBPH terminated early at ~3.06 ms.

Across all 1,804 locked confirmation conditions, 1,156 cases were recognized correctly by both LBPH and SFace, 0 cases were recognized correctly by LBPH alone, 438 cases were recognized correctly by SFace alone, and 210 cases failed both systems. Because LBPH-only correct is exactly 0, P(SFace correct | LBPH correct) = 100.0%. SFace strictly subsumes LBPH across the evaluated transformation space. LBPH does not expand the system's recognition ceiling; rather, it serves as a selective computational shortcut that resolves easy queries in 3.06 ms instead of 8.30 ms.

Workload Severity and Latency Profile. Across degradation tiers, LS-Face achieved consistent speedups: Clean (~3.45 ms, 58.4% latency reduction), Light (6.176 ms, 25.6% reduction), Medium (7.550 ms, 9.0% reduction), and Heavy (7.722 ms, 7.0% reduction). While Direct SFace provides a tighter single-path distribution (p50: 8.206 ms, p95: 9.183 ms, p99: 10.073 ms), LS-Face accepts a modest tail latency penalty on ambiguous inputs (p50: 8.265 ms, p95: 11.704 ms, p99: 12.388 ms) in exchange for a substantial 15.48% reduction in overall mean inference time.

# 5 Discussion

## 5.1 Selective Computation and Elimination of Redundancy

The experimental findings demonstrate that traditional sequential cascades are structurally inefficient because they execute classical descriptors indiscriminately on degraded frames that inevitably require deep-model fallback. Quality-first early bypass eliminates this bottleneck by routing degraded frames directly to SFace, cutting dual inference to 0.26% on LFW.

## 5.2 Orthogonal Computational Optimizations

Quality-first bypass and compact descriptor retuning target distinct cost sources: early bypass eliminates wasted classical extraction on degraded frames, while compact LBPH reduces memory footprint by 43.75% (36 KiB vs 64 KiB) and accelerates classical comparison. As demonstrated by ablation, neither optimization alone surpasses direct SFace, but their combination achieves a 15.48% mean latency reduction.

### 5.3 Computational Shortcuts vs. Accuracy Complementarity

The 100.0% subsumption of LBPH by SFace (1,156 / 1,156) establishes that classical descriptors in hybrid cascades operate strictly as computational shortcuts rather than accuracy complements. In embedded edge systems, reducing mean latency from 8.300 ms to 7.015 ms directly improves energy efficiency and throughput, provided tail latency remains within interactive bounds.

Limitations. The permissive reject boundary tau_reject = 140.13 remained inactive on evaluation workloads (d_max = 72.18), indicating that deployed systems can adopt a streamlined three-branch decision architecture.

## 6 Conclusion

### 6.1 Principal Findings

LS-Face demonstrates that hybrid face recognition cascades can achieve superior average inference efficiency compared to direct deep neural network evaluation without compromising recognition accuracy. By combining quality-guided early bypass routing with a compact retuned LBPH descriptor, LS-Face achieved a 15.48% reduction in mean recognition latency over standalone SFace (38.99% over the baseline cascade) while maintaining exact decision parity across 1,804 locked confirmation conditions.

### 6.2 Architectural Implications

By formalizing classical descriptors as selective computational shortcuts and eliminating dual inference on degraded streams, LS-Face establishes a principled design framework for low-power edge biometrics.

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**Disclosure of Interests.** The authors have no competing interests to declare.

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